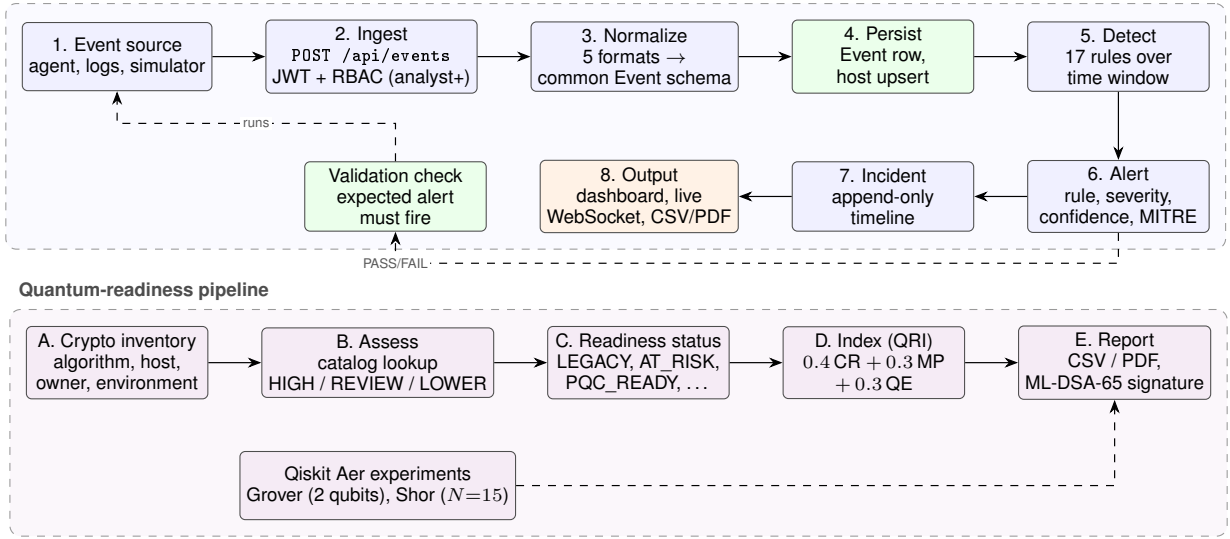
SIEM / SOC detection pipeline
1. Event source
agent, logs, simulator
2. Ingest
POST /api/events
JWT + RBAC (analyst+)
3. Normalize
5 formats → common Event schema
4. Persist
Event row, host upsert
5. Detect
17 rules over time window
6. Alert
rule, severity, confidence, MITRE
7. Incident
append-only timeline
8. Output
dashboard, live WebSocket, CSV/PDF
Validation check
expected alert must fire
runs
PASS/FAIL
Quantum-readiness pipeline
A. Crypto inventory
algorithm, host, owner, environment
B. Assess
catalog lookup
HIGH / REVIEW / LOWER
C. Readiness status
LEGACY, AT_RISK, PQC_READY, …
D. Index (QRI)
0.4 CR + 0.3 MP + 0.3 QE
E. Report
CSV / PDF, ML-DSA-65 signature
Qiskit Aer experiments
Grover (2 qubits), Shor (N=15)